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MAT 213

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Matlab Project 1

1. (a) a = 9, b = 1; these planes are neither parallel nor perpendicular

```
>> X = [9,3,-3]

X =

     9     3    -3

>> Y = [2,-1,-3]

Y =

     2    -1    -3

>> dot(X,Y)

ans =

    24

>> cross(X,Y)

ans =

   -12    21   -15
```

(b) These planes are neither parallel nor perpendicular

```
>> X = [-3,-9,108]

X =

    -3    -9   108

>> Y = [2,6,72]

Y =

     2     6    72

>> dot(X,Y)

ans =

        7716

>> cross(X,Y)

ans =

       -1296         432           0
```

(c) The dot product is 0, so these planes are perpendicular

```
>> X = [9,3,-36]
X =
     9     3   -36
>> Y = [-2,18,1]
Y =
    -2    18     1
>> dot(X,Y)
ans =
     0
>> cross(X,Y)
ans =
   651    63   168
```

2.

```
f =
  function_handle with value:
    @(x,y)9*y^2*sin(x^2*y)
>> diff(f,x)
ans =
18*x*y^3*cos(x^2*y)
>> diff(f,y)
ans =
18*y*sin(x^2*y) + 9*x^2*y^2*cos(x^2*y)
```

```
>> diff(diff(f,y,2),x)
ans =
108*x*y*cos(x^2*y) - 108*x^3*y^2*sin(x^2*y) - 18*x^5*y^3*cos(x^2*y)
>> diff(diff(f,x,2),y)
ans =
54*y^2*cos(x^2*y) - 36*x^4*y^4*cos(x^2*y) - 162*x^2*y^3*sin(x^2*y)
>> diff(diff(f,x,3),y,2)
ans =
72*x^7*y^5*cos(x^2*y) - 2304*x^3*y^3*cos(x^2*y) + 828*x^5*y^4*sin(x^2*y) - 1296*x*y^2*sin(x^2*y)
```

3.

```
>> syms z
>> syms l
>> [l,x,y,z] = solve(y + 2*z - l*y*z, x + 2*z - l*x*z, 2*x + 2*y - l*x*y, -x*y*z + 2916000)

l =

                     1/45
  (3^(1/2)*1i)/90 - 1/90
- (3^(1/2)*1i)/90 - 1/90


x =

                180
- 90 - 3^(1/2)*90i
- 90 + 3^(1/2)*90i


y =

                180
- 90 - 3^(1/2)*90i
- 90 + 3^(1/2)*90i


z =

                 90
- 45 - 3^(1/2)*45i
- 45 + 3^(1/2)*45i
```

```
>> f = @(x,y,z) x*y + 2*x*z + 2*y*z

f =

  function_handle with value:

    @(x,y,z)x*y+2*x*z+2*y*z
```

```
>> f(1,2,1458000)

ans =

     8748002

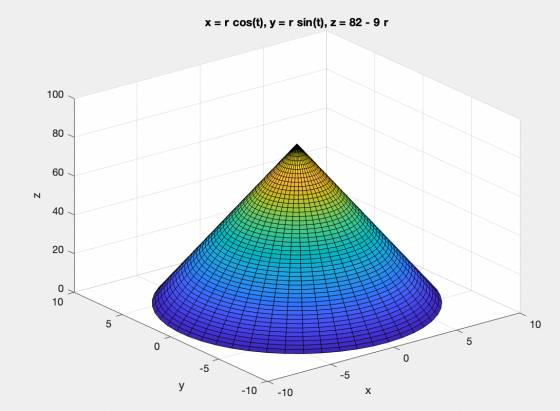
>> f(180,180,90)

ans =

       97200
```

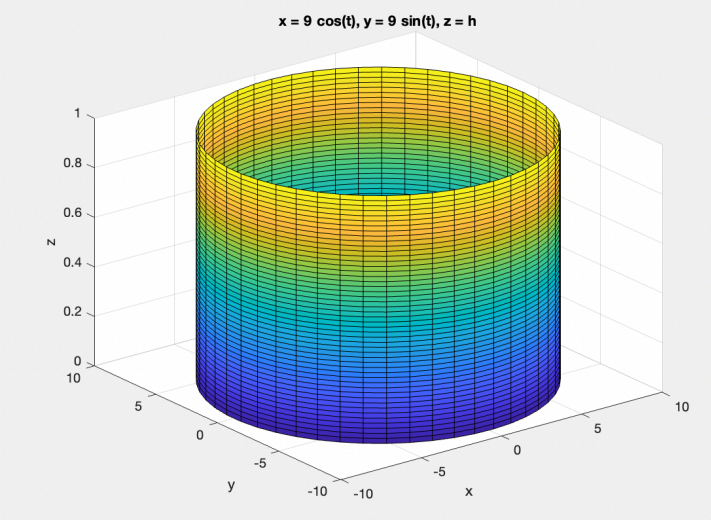
4. (a)

```
>> syms t r
>> ezsurf(r*cos(t),r*sin(t),82-9*r,[0 82/9 0 2*pi])
```



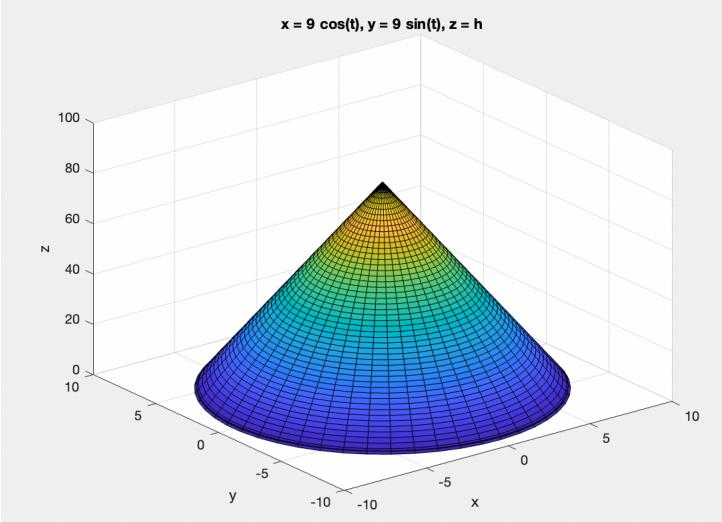
(b)

```
>> syms t h
>> ezsurf(9*cos(t),9*sin(t),h,[0 1 0 2*pi])
```



(c)

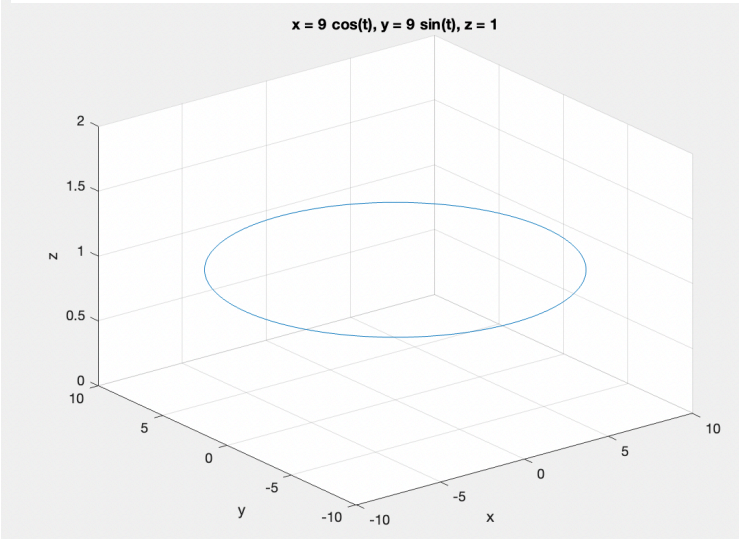
```
>> ezsurf(r*cos(t),r*sin(t),82-9*r,[0 82/9 0 2*pi])
>> hold on
>> ezsurf(9*cos(t),9*sin(t),h,[0 1 0 2*pi])
>> hold off
```



With note that the cylinder is not easily visible because of its extreme height difference against the cone

(d)

```
>> ezplot3(9*cos(t),9*sin(t),1,[0 2*pi])
```



5. (a)

```
>> syms r t
>> int(int((18-r^2-r^2)*r,r,0,3),t,0,2*pi)

ans =

81*pi
```

(b)

```
>> ezsurf(r*cos(t),r*sin(t),r^2,[0 3 0 2*pi])
hold on
ezsurf(r*cos(t),r*sin(t),18-r^2,[0 3 0 2*pi])
hold off
```

